

Savitribai Phule Pune University Second Year of Information Technology (2024 Course) Course Code: PCC-253-ITT Course Name: Probability & Statistics		
Teaching Scheme	Credits	Examination Scheme
Theory : 02 H/W	02	CCE: 30M ESE : 70M
Prerequisite Courses, if any: Set Theory and Logic, Differential and Integral Calculus, Basic Linear Algebra		
Companion Course, if any: Data Science or Machine Learning Foundations, Statistical Inference, Numerical Methods		
Course Objectives: The objective of this course is - To introduce the foundational principles of probability theory. - To study random variables, distributions, and expectations. - To analyze statistical measures including mean, variance, and higher-order moments. - To impart knowledge of sampling theory and estimation techniques. - To develop skills in hypothesis testing for real-world applications.		
Course Outcomes: On completion of the course, learner will be able to – CO1: Apply probability theory to model uncertain systems and events. L3 CO2: Analyze and interpret various probability distributions. L4 CO3: Use moment generating functions and inequalities (Markov and Chebyshev) to analyze random variables. L4 CO4: Apply population parameters using sampling and construct confidence intervals. L3 CO5: Conduct statistical hypothesis tests to support decision-making. L3		
Course Contents		
Unit No: I	Probability Fundamentals and Set Theory	05 Hours
Algebra of sets, Axiomatic, classical, and frequentist definitions of probability, Conditional probability, independence, Bayes' theorem.		
Exemplars /Case Studies	Medical Testing and False Positives Context: Use of conditional probability and Bayes' Theorem in understanding false positives in COVID-19 tests. Learning Outcome: Apply conditional probability and Bayes' Theorem to real-world diagnostic scenarios.	
Reference Books	S.M. Ross , Introduction to Probability and Statistics for Engineers and Scientists – Chapters 1–2 V.K. Rohatgi & A.K. Md. E. Saleh , An Introduction to Probability and Statistics – Chapter 2 H.J. Larson , Introduction to Probability Theory and Statistical Inference – Chapter 1	

Unit No: II	Random Variables and Expectation	07 Hours
Types of random variables (discrete, continuous, mixed), PMF, PDF, CDF, Expectation, variance, higher-order moments, Moment generating functions, Markov and Chebyshev inequalities		
Exemplars/Case Studies	Warranty Analysis in Manufacturing Context: Modeling the life of a product (e.g., a LED bulb) using exponential and normal distributions. Learning Outcome: Calculate expected life, variance, and reliability.	
Reference Books	E.J. Dudewicz & S.N. Mishra , Modern Mathematical Statistics – Chapter 3 V.K. Rohatgi , An Introduction to Probability and Statistics – Chapters 3–4 S.M. Ross , Introduction to Probability and Statistics – Chapters 3–4 Mood, Graybill & Boes , Introduction to the Theory of Statistics – Chapters 2–3	

Unit No: III	Distributions and Functions of Random Variables	08 Hours
Binomial, Poisson, Geometric, Exponential, Normal, Gamma, Beta, Transformation techniques, Joint distributions, marginal and conditional distributions, Covariance, correlation, bivariate normal distribution		
Exemplars/Case Studies	- Traffic Flow Modeling Using Poisson Distribution Context: Modeling the number of vehicles passing a toll gate per minute using Poisson distribution. Learning Outcome: Understand real-world applications of discrete distributions and parameter estimation.	
Reference Books	S.M. Ross – Chapters 5–6 W.W. Hines & D.C. Montgomery, Probability and Statistics in Engineering – Chapter 4 V.K. Rohatgi – Chapter 5 Mood, Graybill & Boes – Chapters 4–5	

Unit No: IV	Sampling and Estimation	05 Hours
Sampling distributions, Central Limit Theorem, Point estimation: unbiasedness, consistency, Maximum likelihood and method of moments, Confidence intervals		
Exemplars/Case Studies	- Market Survey for Launching a New Product Context: A company samples 100 consumers to estimate the average monthly expenditure on a fitness product. Learning Outcome: Construct confidence intervals and evaluate sample bias.	
Reference Books	Dudewicz & Mishra – Chapters 5–6 J.S. Milton & J.C. Arnold, Introduction to Probability and Statistics – Chapters 6–7 Rohatgi & Saleh – Chapters 6–7 Mood, Graybill & Boes – Chapter 6	

Unit No: V	Hypothesis Testing and Decision Theory				05 Hours
Hypothesis formulation, critical regions, Type I & II errors, Neyman-Pearson lemma, Z, t, Chi-square, and F tests, Goodness-of-fit, tests for proportions and variance, Applications in engineering and sciences					
Exemplars/Case Studies		- A/B Testing in E-commerce - Context: Comparing two website versions to determine which leads to better conversion rates using hypothesis testing. - Learning Outcome: Formulate null and alternative hypotheses, conduct z/t-tests, and interpret results.			
Reference Books		- S.M. Ross – Chapter 7 - Mood, Graybill & Boes – Chapters 7–9 - Milton & Arnold – Chapter 9 - W.W. Hines & D.C. Montgomery – Chapter 8			
Learning Resources					
S. N	Book Title	Author(s)	ISBN	Publisher	Edition / Year
1	Introduction to Probability and Statistics for Engineers and Scientists	Sheldon M. Ross	978-0124157822	Academic Press (Elsevier)	5th Edition, 2014
2	An Introduction to Probability and Statistics	V.K. Rohatgi, A.K. Md. E. Saleh	978-1118294405	Wiley	3rd Edition, 2015
3	Modern Mathematical Statistics	E.J. Dudewicz, S.N. Mishra	978-0471030151	Wiley	1st Edition, 1988
4	Introduction to the Theory of Statistics	A.M. Mood, F.A. Graybill, D.C. Boes	978-0070428645	McGraw-Hill	3rd Edition, 1974
5	Introduction to Probability and Statistics	J.S. Milton, J.C. Arnold	978-0079130341	McGraw-Hill Education	4th Edition, 2003
6	Introduction to Probability Theory and Statistical Inference	H.J. Larson	978-0471866453	Wiley	3rd Edition, 1982
7	Probability and Statistics in Engineering	W.W. Hines, D.C. Montgomery, D.M. Goldsman, C.M. Borror	978-0470631472	Wiley	4th Edition, 2003
Ebooks: Introduction to Probability and Statistics for Engineers and Scientists Author(s): Sheldon M. Ross Publisher: Academic Press (Elsevier) Edition/Year: 5th Edition, 2014 ISBN: 978-0123948113 https://minerva.it.manchester.ac.uk/~saralees/statbook3.pdf					

- An Introduction to Probability and Statistics Author(s): Vijay K. Rohatgi, A.K. Md. Ehsanes Saleh, Publisher: Wiley Edition/Year: 2nd Edition, 2000 ISBN: 978-0471326724 https://www.usb.ac.ir/FileStaff/5518_2023-2-8-8-33-53.pdf
- Modern Mathematical Statistics Author(s): Edward J. Dudewicz, Satya N. Mishra Publisher: Wiley Edition/Year: 1st Edition, 1988 ISBN: 978-0471814726 <https://archive.org/details/modernmathematic0000dude/page/n5/mode/2up>

NPTEL / SWAYAM Video Lectures

NPTEL - Probability and Statistics by Prof. H.K. Kesavan (IIT Madras)

NPTEL - Probability and Statistics by Prof. Somesh Kumar (IIT Kharagpur)

SWAYAM – Fundamentals of Probability and Statistics